

Continuation Versus Temporary Interruption of Immunomodulatory Agents During Infections in Patients With Inflammatory Rheumatic Diseases: A Randomized Controlled Trial

Merel A. A. Opdam,^{1,2} Nathan den Broeder,^{1,2} Reinout van Crevel,^{2,3} Lisa Schapink,^{1,2} Léon Raymakers,^{1,2} Jasper Broen,^{3,4} Lise M. Verhoef,^{1,2} and Alfons A. den Broeder^{1,4,5}

¹Department of Rheumatology, Sint Maartenskliniek, Nijmegen, The Netherlands; ²Department of Internal Medicine and Radboud Community for Infectious Diseases, Radboud University Medical Center, Nijmegen, The Netherlands; ³Regional Rheumatology Centre, Máxima Medical Centre, Eindhoven, The Netherlands; and ⁴Department of Rheumatology, Radboudumc, Nijmegen, The Netherlands

Background. Immunomodulatory agents (IA) are commonly used to treat inflammatory rheumatic diseases (IRDs). Guidelines typically recommend temporarily interrupting IA during infections, but evidence supporting this approach is limited and continuation may be equally safe or even preferable. The objective of this study was to compare these 2 treatment strategies.

Methods. We conducted a large multi-center, open-label, randomized controlled trial (Dutch Trial Register (NL8922), in which infection-free IRD patients on IA were randomized 1:1 to either continue or temporarily interrupt IA treatment when experiencing their first clinically relevant infection. Primary outcome was the proportion of patients with a serious infection (requiring hospitalization or intravenous treatment), analyzed in the modified intention-to-treat (mITT) population using the Cochran-Mantel-Haenszel method. The mITT population included all patients who experienced a clinically relevant infection during follow-up. Secondary analyses included Complier Average Causal Effect (CACE) models to account for non-adherence.

Results. Of 1142 patients enrolled (1667 patient years of follow-up), 474 developed a clinically relevant infection (mITT population). Serious infections occurred in 12 of 233 patients (5.15%) in the interruption group and in 9 of 241 (3.73%) in the continuation group, adjusted risk difference was 1.71% (95% CI −1.99 to 5.39). CACE analysis showed a risk difference of 4.51 (95% confidence interval [CI]: −7.32 to 16.34) in favor of continuation.

Conclusions. These findings suggest that temporary interruption and continuation of IA result in similar risk and outcome of infections in IRD patients. Although this study is limited by low statistical power, the findings suggest that continuation of IA during an infection is safe.

Keywords. inflammatory rheumatic diseases; infections; immunomodulatory agents; randomized controlled trial.

Immunomodulatory agents (IA) are widely used to treat patients with immune mediated inflammatory diseases (IMIDs), including inflammatory rheumatic diseases (IRDs) such as rheumatoid arthritis (RA), psoriatic arthritis (PsA), and axial spondyloarthritis (axSpA) [1, 2]. Even though IA are effective and safe, IRD patients using IA may be more susceptible to severe infections [3–5]. Moreover, it has been shown that infection risk is already higher in specific IRDs such as RA, and

that higher disease activity is associated with higher infection risk. Other patient characteristics such as age and comorbidities can add to a higher risk of (severe) infection [6–8]. Therefore, it is of importance to gain knowledge on how to manage infections in IRD patients using IA.

It is unclear whether IA should be temporarily interrupted or continued in case of an infection. For most IA, clinical guidelines and the Summary of Product Characteristics (SmPC) advise to temporarily interrupt treatment in case of a (severe) intercurrent infection, mostly based on infection management as done in registration trials [9, 10]. Theoretically, temporary cessation of the treatment will lead to a lower drug level, putatively resulting in a reduced immunosuppressive effect and prevention of further escalation of the infection. However, evidence on this topic is very scarce and there are some arguments in favor of continuation of IA in case of an infection.

First, the reduction in IA drug levels may be reached too late to impact the course of infection, especially as many IA have long half-lives. Second, interruption of the IA might evoke a

Received 25 March 2025; editorial decision 13 July 2025; published online 12 August 2025

Correspondence: M. A. A. Opdam, Department of Rheumatology, Sint Maartenskliniek, Hengstdal 3, 6574 NA Gelderland, Nijmegen, The Netherlands (m.opdam@maartenskliniek.nl); A. A. den Broeder, Department of Rheumatology, Sint Maartenskliniek, Hengstal 3, 6574 NA Gelderland, Nijmegen, The Netherlands (a.denbroeder@maartenskliniek.nl).

Clinical Infectious Diseases® 2026;82(3):420–6

© The Author(s) 2025. Published by Oxford University Press on behalf of Infectious Diseases Society of America. All rights reserved. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.
https://doi.org/10.1093/cid/ciaf442

disease flare with subsequent increased infection risk. Increased disease activity could also result in need for extra glucocorticoid use, which is associated with increased infection risks as well. The most important, and perhaps somewhat counterintuitive argument, is that continuation of IA could result in better infection outcomes. This is because negative infection outcomes are often driven by an excessive host response which can be countered by IA, as seen in sepsis or severe coronavirus disease 2019 (COVID-19) [11–14] and because some IA have innate anti-infective or anti-viral properties themselves [15–17].

To our knowledge, there are no clinical trials or observational studies directly comparing the outcomes of continuation of IA with interruption of IA during infection. The hypothesis that continuing IA treatment may be beneficial in case of infection has only support by indirect and circumstantial evidence. We therefore carried out a randomized controlled trial to assess the effect of continuation of IA versus temporary interruption in case of an incident infection on infection outcome, notably progression to serious infection.

METHODS

Trial Design and Oversight

This study was a large, pragmatic, exploratory, open-label, multicenter, randomized controlled strategy study among IRD patients using (one or multiple) IA in any dose. The trial was conducted in 5 locations of 2 large medical centers in The Netherlands (Sint Maartenskliniek and Máxima Medical Centre) in accordance with the principles of the Declaration of Helsinki, Dutch legislation and guidelines, and Good Clinical Practice. Ethical approval was granted by the local Medical Research Ethics Committee (MREC) (file number 2020-6444). All patients provided written informed consent. This trial was funded by the Sint Maartenskliniek ([R]SMK). In the supplementary files, additional information on the methods of this study, the most recent trial protocol, overview of protocol amendments (Supplementary Table 4) and the CONSORT checklist are provided. An overview of enrolment, interventions and assessments is shown in Supplementary Figure 1.

Participants and Eligibility Criteria

In order to maximize external validity, inclusion criteria for this pragmatic trial were as non-restrictive as possible. Patients were eligible for inclusion if they (1) had a clinical diagnosis of one of the following IRDs: rheumatoid arthritis (RA), psoriatic arthritis (PsA) or axial spondyloarthritis (axSpA), (2) were 16 years or older, (3) used one or more IA from Supplementary Table 1, (4) did not have a clinical infection at time of inclusion and (5) were able to read and communicate in Dutch.

Initial anticipated follow-up of participants was 12 months. Changes in follow-up periods were made due to lagging inclusion rates and lower than expected infection rates, and

increases in follow-up time were made to compensate for this with more patient years follow-up.

Treatment

Participants were randomized 1:1 to IA continuation (intervention) or temporary interruption of IA during an infection (control). Participants were randomised via online database Castor EDC[®] using block randomization with varying block sizes of 6, 8, and 10 and a computer-generated list stratified for use of tumour necrosis factor inhibitors (TNFi), use of concomitant glucocorticoids and risk for severe COVID-19, based on the categorization of the Centers for Disease Control and Prevention (CDC) (see Supplementary File 1 and Supplementary Table 2) [18]. Randomization was concealed until allocation. After randomization, allocation was open-label and participants received information on their allocation with detailed instruction on when to contact the research team and how to act in case of an infection. Treating physicians were encouraged to act according to the participant's allocation in case of an infection. However, all final treatment decisions were at the discretion of the treating physician. This strategy closely mirrors what is done in clinical care concerning infection risk in IA using IRD patients, maximizing generalizability and acceptability.

The allocation only applied to the first clinically relevant infection ($\geq$ grade 2 according to the Common Terminology Criteria for Adverse Events [CTCAE] version 5.0) that occurred in each patient during the study period [19]. If participants experienced more than 1 clinically relevant infection during the study, decisions on continuation or temporary interruption were left to the discretion of the treating physician and the patient (ie, usual care). If IA was added or switched during the study period, the allocation applied to the new IA as well (if included in Supplementary Table 1). If participants discontinued IA treatment, they remained in the study and data was collected as planned. They could (re)start their IA during the study period, enabling them to act upon their allocation. Participants were free to use all other medication as needed.

A more extensive explanation of both treatment strategies is provided in Supplementary File 1.

Outcomes

Primary outcome was the proportion of patients with a serious infection ($\geq$ grade 3 according to the CTCAE version 5.0). A global overview of severity assessment is shown in Table 1. Secondary outcomes included infection characteristics (including severity, duration, and type), (serious) adverse events (SAEs), and disease activity.

Statistical Analysis

Because this study is an exploratory trial, no null and alternative hypotheses were stated. The outcome of the study is the point estimate of the between group difference in proportion of patients with infection. We did perform a sample size

Table 1. Infection Severity

Severity	Explanation
Grade 1	Mild; asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated
Grade 2	Moderate; minimal, local or non-invasive intervention indicated; limiting age-appropriate instrumental ADL ^a
Grade 3	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care ADL ^b
Grade 4	Life-threatening consequences, urgent intervention indicated
Grade 5	Death related to AE

Abbreviations: ADL, activities of daily living; AE, adverse event.

^aInstrumental ADL refer to preparing meals, shopping for groceries or clothes, using the telephone, managing money etc.

^bSelf-care ADL refer to bathing, dressing and undressing, feeding self, using the toilet and taking medications.

calculation to get insight into the precision of estimates that could be obtained. To be able to show a difference of 2.5% or more (patients with a serious infection) between the control and intervention groups (expected 1-year incidence rates of 5.0% and 2.5%), with a 2-sided alpha of 0.05, power of 80%, and 1:1 randomization, a total of 1812 evaluable participants would be needed (see [Supplementary File 1](#)). Accounting for 20% drop-out and non-adherence, the aim was to include 2200 patients.

The primary analysis was done on the modified intention to treat (mITT) population. In this study, the mITT population consisted of all randomised patients who experienced at least 1 clinically relevant ($\geq$ grade 2) infection. The proportion of patients with a serious infection in both treatment groups was compared using the Cochran-Mantel-Haenszel risk difference and confidence interval, adjusting for stratification factors [20].

To assess the effect of continuation versus temporary interruption of IA treatment in several subsets of the study population, the proportion of patients with a serious infection in the subgroups was assessed using the Cochran-Mantel-Haenszel risk difference and confidence interval, adjusting for stratification factors. Statistical analysis were done in STATA/IC 17 for Windows and RStudio.

Additionally, to assess the efficacy of treatment received (taking into account non-compliance), we used the Complier Average Causal Effect (CACE) method as a per protocol analysis (further explained in [Supplementary File 1](#)) [21, 22]. This method allows estimation of the effect in patients who complied to their allocation. This analysis was done on the mITT population.

RESULTS

Participant Characteristics

Between October 2020 and July 2023, 1142 patients with a total follow-up time of 1667 patient years were included. The trial was stopped prematurely because the inclusion in the

recruiting hospitals stalled, and the limits of funding were reached. In total, 569 participants were allocated to the control group (temporary interruption) and 573 to the intervention group (continuation).

A total of 474 patients (233 temporary interruption, 241 continuation) experienced at least 1 clinically relevant infection during follow-up and were included in the mITT population. Patient, disease, and treatment characteristics are shown in [Table 2](#) and are similar between the 2 groups, except for history of previous severe infections and use of oral glucocorticoids in the mITT population. A slight majority of the participants was female, the most common IRD rheumatoid arthritis (around 61%) and 12% to 18.5% had previously experienced a serious infection with slightly different numbers between the entire and mITT population.

A total of 51 participants (4.5%) dropped out of the study before completion of follow-up. Also, 34 of the 51 dropped out participants did not experience an infection before their termination of the study (see [Supplementary File 2](#)).

Infections

In total, 1952 infections occurred during 1667 patient years of follow-up (incidence rate 117 per 100 patient-years, 95% CI: 112 to 122). [Table 3](#) shows the severity assessments of all infections that occurred during the study. Most infections (97.8%) were mild to moderate (grade 1 or 2). Most serious infections were grade 3 infections, for example, leading to hospitalization or intravenous treatment but not to urgent interventions such as admission to the intensive care unit or death. Respiratory tract infections, including COVID-19, were most common (65.1%) as shown in [Table 4](#).

Continuation Versus Temporary Interruption

In the mITT population, serious infections occurred in 12 out of 233 patients (5.15%) in the temporary interruption group and in 9 out of 241 patients (3.73%) in the continuation group, resulting in an unadjusted risk difference of 1.42% (95% CI: -2.30 to 5.13) and an adjusted risk difference of 1.71% (95% CI: -1.99 to 5.39) in favor of continuation ([Figure 1](#)). Subanalyses in different subsets of the study population showed similar adjusted risk differences ([Figure 2](#)) see supplementary file for a more extensive overview of all subgroups and results of subanalyses ([Supplementary Table 3](#)). Median duration of infection was comparable between continuation (11 days [interquartile range {IQR} 7–19]) and temporary interruption (12 days (IQR 7–21), $P = .63$).

With regard to protocol compliance, in the temporary interruption group 106/233 (45.5%) interrupted their IA during their first clinically relevant infection. In the continuation group, 207/241 (85.9%) continued their IA. Based on CACE method, the estimated percentage of compliers is 31.4% (see [Supplementary File 2](#) for further explanation). Results of the

Table 2. Patient, Disease and Treatment Characteristics

	Full Study Population		mITT Population (Patients With at Least 1 Clinically Relevant Infection)	
	Temporary Interruption (n = 569)	Continuation (n = 573)	Temporary Interruption (n = 233)	Continuation (n = 241)
Age years, mean (SD)	60.5 (12.6)	60.1 (12.8)	61.3 (12.4) ^a	59.7 (14.2) ^a
Sex female, n (%)	328 (57.6)	313 (54.6)	149 (63.9)	153 (63.5)
IRD, n (%)				
Rheumatoid arthritis	349 (61.3)	348 (60.7)	140 (60.1)	147 (61.0)
Psoriatic arthritis	170 (29.9)	164 (28.6)	72 (30.9)	67 (27.8)
Spondyloarthritis	50 (8.79)	61 (10.6)	21 (9.01)	27 (11.2)
Disease duration, years, median (IQR)	8 (3–16)	9 (3–18)	10 (4–18) ^a	10 (4–18) ^a
Patient reported disease activity ^b , median (IQR)	2.5 (1.0–5.5)	3.0 (1.0–5.0)	3.0 (1.0–6.0)	3.0 (1.5–6.0)
BMI kg/m ² , median (IQR)	26.0 (23.5–28.7)	25.5 (23.3–28.6)	25.7 (23.5–28.6)	25.6 (23.3–29.4)
History of previous severe infection, n (%) ^c	85 (14.9)	71 (12.4)	43 (18.5)	29 (12.0)
DMARD use, n (%)				
csDMARDs	434 (76.3)	417 (72.8)	166 (71.2) ^a	165 (68.5) ^a
bDMARDs	214 (37.6)	224 (39.1)	105 (45.1) ^a	107 (44.4) ^a
tsDMARDs	32 (5.62)	30 (5.24)	21 (9.01) ^a	21 (8.71) ^a
Oral glucocorticoid use, n (%)	28 (4.92)	33 (5.76)	12 (5.15) ^a	19 (7.88) ^a
Follow-up time, years, median (IQR)	1.46 (1.02–1.87)	1.63 (1.03–1.88)	0.63 (0.24–1.11) ^d	0.67 (0.30–1.12) ^d

Data are presented at baseline unless stated otherwise.

Abbreviations: BMI, body mass index; bDMARDs, biologic synthetic DMARDs; csDMARDs, conventional synthetic DMARDs; DMARDs, disease modifying antirheumatic drugs; IQR, interquartile range; IRD, inflammatory rheumatic disease; mITT, modified intention-to-treat; tsDMARDs, targeted synthetic DMARDs.

^aAt date of infection.

^bOn a scale of 0–10 (0 = no disease activity, 10 = maximum disease activity), reported by patients.

^cReported in electronic patient files.

^dFrom inclusion until date of infection.

Table 3. Infections and Severity

Severity	Frequency (n, %)	Incidence Rate (n/100 Patient-Years)
Grade 1 or 2	1909 (97.8%)	114 (95% CI 109 to 120)
Grade 3	40 (2.1%)	2.6 (95% CI 1.9 to 3.5)
Grade 4	2 (0.1%)	
Grade 5	1 (0.05%)	

Abbreviation: CI, confidence interval.

Table 4. Infection Types

Infection Type	Frequency (n, %)
Respiratory infections (excluding COVID-19)	781 (40.0%)
Infections caused by SARS-CoV-2	489 (25.1%)
Skin infections	211 (10.8)
Urogenital infections	208 (10.7)
Ear, nose, throat, teeth, jaw and maxillofacial infections	103 (5.3)
Abdominal infections	85 (4.4)
Eye infections	40 (2.1)
Other ^a	26 (1.1)
Musculoskeletal infections	9 (0.5)

Abbreviations: COVID-19, coronavirus disease 2019; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.

^aIncluding unspecified infections (eg, general malaise and fever without specific symptoms, Lyme disease).

CACE method showed an unadjusted risk difference in compliers of 4.51% in favor of continuation (95% CI[†] –7.32 to 16.34). Non-compliance was partially unintentional due to inability to

interrupt medication with long treatment intervals. For example, adalimumab is usually administered once every 2 weeks. If an infection started and resolved between 2 administrations, a patient allocated to temporary interrupted could not interrupt their IA and was considered to have continued their IA for the CACE analyses.

Safety

Serious Adverse Events

In total, 178 SAEs, of which 135 non-infectious, were reported during the study. Seventy SAEs occurred in the temporary interruption group, 65 in the continuation group (Tables 4 and 5).

Disease Activity

The first patient-reported disease activity after the first clinically relevant infection was similar between both treatment groups. Median disease activity was 3.5 (IQR 2.0–7.0) for temporary interruption and 4.0 (IQR 2.0–6.0) for continuation ($P = .96$).

DISCUSSION

This is the first study comparing two frequently used strategies when confronted with an infection in IRD patients using IA: continuation versus temporary interruption of the drug. Although this was not a formal hypothesis testing trial, and the trial was underpowered, no large signal of better infection outcomes was found for temporary interruption versus

Table 5. Overview of Serious Adverse Events in Both Groups

Type of SAE n (%)	Temporary Interruption (n = 569)	Continuation (n = 573)	Risk Difference and 95% CI
Cardiac disorders	16 (2.81%)	14 (2.44%)	0.37% (−1.49 to 2.23)
Neoplasms	13 (2.28%)	9 (1.57%)	0.71% (−0.88 to 2.30)
Gastrointestinal disorders	7 (1.23%)	8 (1.40%)	−0.17% (−1.50 to 1.16)
Injury, poisoning, and procedural complications	8 (1.41%)	6 (1.22%)	0.36% (−1.03 to 1.76)
Nervous system disorders	7 (1.23%)	7 (1.22%)	0.01% (−1.32 to 1.34)
Vascular disorders	3 (0.53%)	8 (1.40%)	−0.87% (−1.97 to 0.23)
General disorders and administration site conditions	4 (0.70%)	2 (0.35%)	0.35% (−0.49 to 1.20)
Respiratory, thoracic and mediastinal disorders	4 (0.70%)	2 (0.35%)	0.35% (−0.49 to 1.20)
Renal and urinary disorders	2 (0.35%)	3 (0.52%)	−0.17% (−0.86 to 0.52)
Other ^a	6 (1.05%)	6 (1.05%)	0.00% (−1.18 to 1.18)
Serious infections	26 (4.57%)	17 (2.97%)	1.60% (−0.61 to 3.81)

Abbreviations: CI, confidence interval; SAE, severe adverse event.

^aIncluding eye disorders, immune system disorders, blood and lymphatic disorders, investigations, metabolism and nutritional disorders, musculoskeletal and connective tissue disease disorders, pregnancy, puerperium and perinatal conditions, psychiatric disorders and reproductive system and breast disorders.

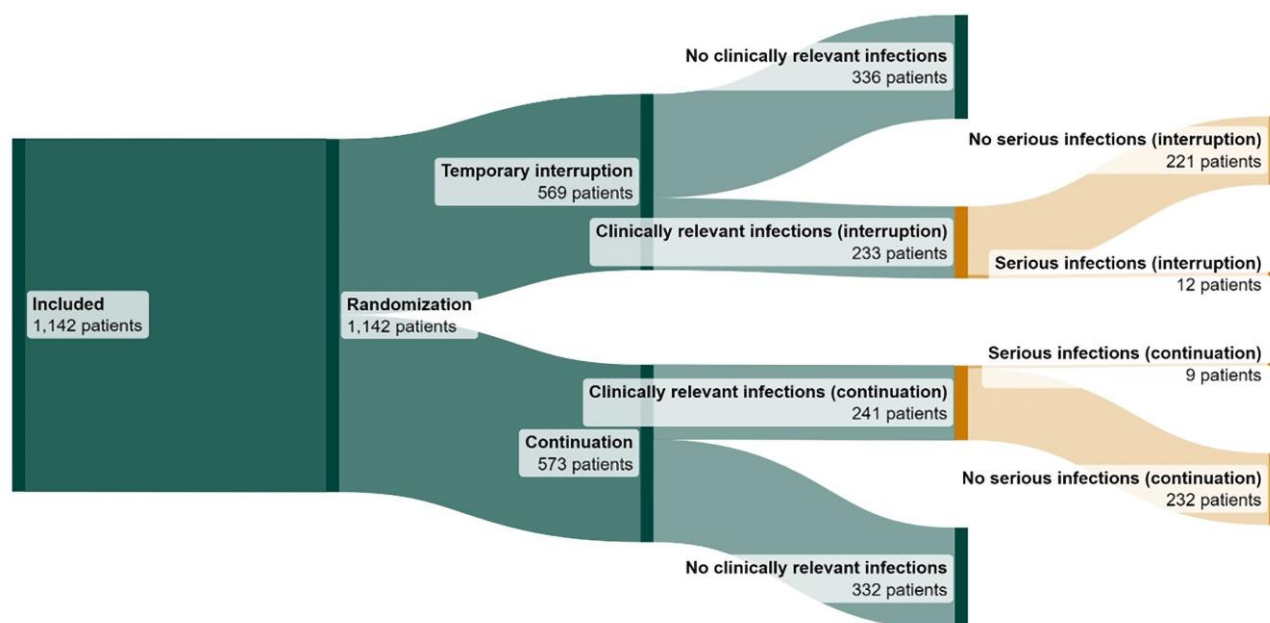


Figure 1. Flow diagram from included patients to patients with serious infection. This flow diagram was made using [SankeyMATIC.com](https://sankeymatic.com/).

continuation of IA in case of a clinically relevant infection. These findings suggest that continuation of IA in case of an intercurrent infection might be a safe option. Subgroup analyses and an estimation of the treatment effect in compliers using the CACE method were in line with the main findings of this study, although less precise.

Circumstantial evidence supports the suggestion that continuation of IA during infections is a safe and potentially better option. In the first place, several IA interfere in cytokine pathways that are involved in inflammation and can be used to treat hyperinflammation in sepsis or septic shock. Two systematic reviews on the use of TNFi during sepsis showed lower mortality rates in septic patients receiving TNFi compared to placebo [11, 12]. Moreover,

an observational study found a lower risk of fatal outcome of sepsis in IRD patients treated with biological disease modifying antirheumatic drugs (bDMARDs) compared to conventional synthetic (cs)DMARDs [23]. Similarly, several IA have been studied as potential treatment for (severe) Coronavirus Disease 2019 (COVID-19) and currently corticosteroids, the IL-6 receptor blockers (IL-6R-blockers) tocilizumab and sarilumab, and JAK inhibitors (JAKi) baricitinib are recommended by the World Health Organization (WHO) as treatment for severe COVID-19 [13, 14]. Additionally, a randomized trial on perioperative continuation or temporary interruption of methotrexate on postoperative infection risk after elective orthopaedic surgery showed much lower infection rates in patients who continued methotrexate compared to

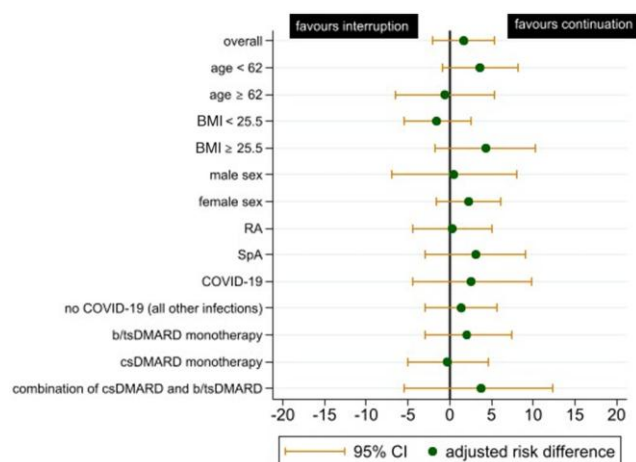


Figure 2. Adjusted risk differences in different subsets of the population with 95% confidence interval. Abbreviations: BMI, body mass index; COVID-19, coronavirus disease 2019; DMARD, disease modifying antirheumatic drug; RA, rheumatoid arthritis; SpA, spondyloarthritis (including psoriatic arthritis and axial spondyloarthritis).

patients who interrupted methotrexate [24]. With regard to bDMARDs, data on the best perioperative treatment strategy are inconsistent and inconclusive [24].

Strengths of this study include the prospective and randomized design. A specific aspect of this study is that, for logistical and generalisability reasons, participants were randomized to one of the strategies before an infection had occurred. However, we do not expect this to have impacted our results as patient characteristics were similar at both baseline and moment of first clinically relevant infection. Also, our randomization strategy mirrors what is done in clinical practice (counselling and instructing patients according to a certain treatment plan on what to do before an infection has actually occurred). With regard to generalisability, we aimed to include a study population that closely mirrors the “real-world” population by using as non-restrictive eligibility criteria as possible. Many studies within the field of IRDs focus on RA, but we also included patients with axSpA and PsA, improving applicability of the results. Infection incidence rates were comparable or higher in our study compared to other studies, this might be explained by the detailed documentation of infections using monthly questionnaires, reporting all infections including common colds, and collection of data during the COVID-19 pandemic as 25% of all infections were caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) [25–28]. Of note, the serious infection rate of 2.6 per 100 patient-years is in full accordance with the ample evidence base on this topic [4, 29, 30].

A limitation of this study is the still limited precision—in spite of the large number of included patients—due to serious infections still being rare. Additionally, we included fewer patients than needed according to our sample size calculation

due to fewer centres and departments participating than initially anticipated and limitations regarding funding, although we in part were able to compensate that by extending exposure time per patient (including 1667 patient-years of follow-up, with an aimed sample size of 2200 patients being followed for 1 year). Additional post hoc power calculations show that with 1142 patients included, an expected incidence rate of serious infections of 2.5% and 5% and 2-sided alpha of 0.05, power is 60.4%. When using the patient years included (841 patient years in the continuation group, 826 patient years in the interruption group), this results in a power of 76.7%. Nevertheless, we did not find a large signal of better infection outcomes for temporary interruption versus continuation. Also, (progression to) serious infection being rare is an important finding, as it results in very large numbers needed for any strategy to be superior. Another limitation was the compliance being lower than expected. Many people allocated to temporary interruption continued their IA, resulting in less contrast between the two treatment groups. This was partly due to unintentional non-compliance, as IA with larger treatment intervals cannot always be interrupted during an infection. However, participants also often did not remember to interrupt or indeed preferred to continue their IA during an infection despite clear instructions. Previous qualitative research showed that various patient related factors, including fear of disease flare, infection severity and instructions received from a health care provider, influence the decision to interrupt or continue IA treatment, and indeed we also expect this to happen in clinical practice [31]. To take into account this non-compliance, we did a secondary analysis using the CACE method, which allows estimation of the treatment effect in compliers only. The results of this method are less precise but also reinforce that temporary interruption has no clear benefits large continuation.

In conclusion, the data derived from this exploratory trial do not suggest a large clinically relevant benefit of temporary interruption over continuation. However, the relatively low number of serious infections limits the statistical power of this study. In clinical practice, shared decision making remains essential and physicians and patients should discuss both options and weigh the possible (dis)advantages of both treatment strategies.

Supplementary Data

Supplementary materials are available at [Clinical Infectious Diseases](https://academic.oup.com/cid/article/82/3/420/8232734) online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

Author contributions. All authors meet the ICMJE criteria for authorship. N. d. B., R. v. C., L. S., L. R., L. V., and A. d. B. were responsible for the concept and design of the study. M. O., J. B., L. V., and A. d. B. contributed to data collection. M. O., N. d. B., L. V., and A. d. B. contributed to

analysis of the data. M. O. and N. d. B. have directly accessed and verified the underlying data reported in the manuscript. M. O. wrote the first draft of this manuscript, which was critically reviewed by all other authors (N. d. B., R. v. C., L. S., L. R., J. B., L. V., A. d. B.). All authors had full access to all the data in the study. All authors have approved the final version of this manuscript and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Acknowledgments. The authors thank all patients who participated in the trial, all student assistants who contributed to conducting this trial, and Dr Stuart Baker for guidance and advice on the CACE analysis.

Data availability. Individual de-identified participant data will be made available to researchers who provide a methodologically sound proposal to achieve aims in the approved proposal or for individual participant data meta-analyses. Data requestors will need to sign a data access agreement.

Financial support. This trial was funded by the Sint Maartenskliniek ([R]SMK). The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report

Potential conflicts of interest. A. d. B. reports grants for research and quality of care projects to the institution from Lilly, Abbvie, Galapagos, Novartis, Pfizer, Gilead, Sanofi, Biogen, Celltrion, UCB. All other authors report no potential conflicts.

All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

References

- Gossec L, Baraliakos X, Kerschbaumer A, et al. EULAR recommendations for the management of psoriatic arthritis with pharmacological therapies: 2019 update. *Ann Rheum Dis* **2020**; 79:700–12.
- Smolen JS, Landewé RBM, Bergstra SA, et al. EULAR recommendations for the management of rheumatoid arthritis with synthetic and biological disease-modifying antirheumatic drugs: 2022 update. *Ann Rheum Dis* **2023**; 82:3–18.
- Listing J, Gerhold K, Zink A. The risk of infections associated with rheumatoid arthritis, with its comorbidity and treatment. *Rheumatology (Oxford)* **2013**; 52: 53–61.
- Singh JA, Cameron C, Noorbaloochi S, et al. Risk of serious infection in biological treatment of patients with rheumatoid arthritis: a systematic review and meta-analysis. *Lancet* **2015**; 386:258–65.
- Riley TR, George MD. Risk for infections with glucocorticoids and DMARDs in patients with rheumatoid arthritis. *RMD Open* **2021**; 7:e001235.
- Accortt NA, Lesperance T, Liu M, et al. Impact of sustained remission on the risk of serious infection in patients with rheumatoid arthritis. *Arthritis Care Res (Hoboken)* **2018**; 70:679–84.
- Au K, Reed G, Curtis JR, et al. High disease activity is associated with an increased risk of infection in patients with rheumatoid arthritis. *Ann Rheum Dis* **2011**; 70: 785–91.
- Hasseli R, Mueller-Ladner U, Hoyer BF, et al. Older age, comorbidity, glucocorticoid use and disease activity are risk factors for COVID-19 hospitalisation in patients with inflammatory rheumatic and musculoskeletal diseases. *RMD Open* **2021**; 7:e001464.
- Landewé RBM, Kroon FPB, Alunno A, et al. EULAR recommendations for the management and vaccination of people with rheumatic and musculoskeletal diseases in the context of SARS-CoV-2: the November 2021 update. *Ann Rheum Dis* **2022**; 81:1628–39.
- Biologicals. **2013**. Available at: https://richtlijnendatabase.nl/richtlijn/biologische_startpagina_-_biologische_dmard_s_en_targeted_synthetic_dmard_s.html. Accessed 11 March 2024.
- lv S, Han M, Yi R, Kwon S, Dai C, Wang R. Anti-TNF- α therapy for patients with sepsis: a systematic meta-analysis. *Int J Clin Pract* **2014**; 68:520–8.
- Qiu P, Cui X, Sun J, Welsh J, Natanson C, Eichacker PQ. Antitumor necrosis factor therapy is associated with improved survival in clinical sepsis trials: a meta-analysis. *Crit Care Med* **2013**; 41:2419–29.
- Alunno A, Najm A, Machado PM, et al. 2021 update of the EULAR points to consider on the use of immunomodulatory therapies in COVID-19. *Ann Rheum Dis* **2022**; 81:34–40.
- Agarwal A, Hunt BJ, Stegemann M, et al. A living WHO guideline on drugs for covid-19. *BMJ* **2020**; 370:m3379.
- Ben-Zvi I, Kivity S, Langevitz P, Shoenfeld Y. Hydroxychloroquine: from malaria to autoimmunity. *Clin Rev Allergy Immunol* **2012**; 42:145–53.
- Altenburg J, de Graaff CS, van der Werf TS, Boersma WG. Immunomodulatory effects of macrolide antibiotics—part 1: biological mechanisms. *Respiration* **2010**; 81:67–74.
- Labro MT. Immunomodulatory effects of antimicrobial agents. Part II: antiparasitic and antifungal agents. *Expert Rev Anti Infect Ther* **2012**; 10:341–57.
- CDC updates, expands list of people at risk of severe COVID-19 illness. **2020**. Available at: <https://archive.cdc.gov/#/details?q=Evidence%20used%20to%20update%20the%20list%20of%20underlying%20medical%20conditions%20that%20increase%20a%20person's%20risk%20of%20severe%20illness%20from%20COVID-19&start=0&rows=10&url=https://www.cdc.gov/media/releases/2020/p0625-update-expands-covid-19.html>. Accessed 10 April 2024.
- Services USDoHaH. Common terminology criteria for adverse events (CTCAE) version 5.0. 2017. Available at: https://ctep.cancer.gov/protocoldevelopment/electronic_applications/docs/ctcae_v5_quick_reference_5x7.pdf. Accessed 11 March 2024.
- Klingenberg B. A new and improved confidence interval for the Mantel-Haenszel risk difference. *Stat Med* **2014**; 33:2968–83.
- Baker SG, Kramer BS, Lindeman KS. Latent class instrumental variables: a clinical and biostatistical perspective. *Stat Med* **2016**; 35:147–60.
- Imbens G, Angrist J. Identification and estimation of local average treatment effects. *Econometrica* **1994**; 62:467–76.
- Richter A, Listing J, Schneider M, et al. Impact of treatment with biologic DMARDs on the risk of sepsis or mortality after serious infection in patients with rheumatoid arthritis. *Ann Rheum Dis* **2016**; 75:1667–73.
- Grennan DM, Gray J, Loudon J, Fear S. Methotrexate and early postoperative complications in patients with rheumatoid arthritis undergoing elective orthopaedic surgery. *Ann Rheum Dis* **2001**; 60:214–7.
- Gron KL, Glinthorp B, Nørgaard M, et al. Overall infection risk in rheumatoid arthritis during treatment with Abatacept, rituximab and tocilizumab; an observational cohort study. *Rheumatology* **2020**; 59:1949–56.
- Doran MF, Crowson CS, Pond GR, O'Fallon WM, Gabriel SE. Frequency of infection in patients with rheumatoid arthritis compared with controls: a population-based study. *Arthritis Rheum* **2002**; 46:2287–93.
- Germano V, Cattaruzza MS, Osborn J, et al. Infection risk in rheumatoid arthritis and Spondyloarthritis patients under treatment with DMARDs, corticosteroids and TNF- α antagonists. *J Transl Med* **2014**; 12:77.
- Greenberg JD, Reed G, Kremer JM, et al. Association of methotrexate and tumour necrosis factor antagonists with risk of infectious outcomes including opportunistic infections in the CORRONA registry. *Ann Rheum Dis* **2010**; 69:380–6.
- Dartel SAAV, Fransen J, Kievit W, et al. Difference in the risk of serious infections in patients with rheumatoid arthritis treated with Adalimumab, infliximab and etanercept: results from the Dutch rheumatoid arthritis monitoring (DREAM) registry. *Ann Rheum Dis* **2013**; 72:895–900.
- Bechman K, Subesinghe S, Norton S, et al. A systematic review and meta-analysis of infection risk with small molecule JAK inhibitors in rheumatoid arthritis. *Rheumatology* **2019**; 58:1755–66.
- Opdam MAA, Vriezeekolk JE, Broen J, den Broeder AA, Verhoef LM. Exploring the perspective of patients with immune-mediated inflammatory diseases and care providers on the use of immunomodulatory drugs in infections: an interview study. *Rheumatol Adv Pract* **2023**; 7:rkad003.